Paleomagnetism of the Keweenawan Chipman Lake and Seabrook Lake carbonatite complexes, Ontario[1]

D. T. A. Symons
Department of Geology, University of Windsor, Windsor, Ont., Canada N9B 3P4

Received October 28, 1991
Revision accepted January 23, 1992

The Chipman Lake complex crops out as a series of carbonatite and related alkalic mafic dikes in the Wabigoon Subprovince of the Superior Province, whereas the Seabrook Lake complex crops out as an alkalic syenite – carbonatite stock in the Abitibi Subprovince. Paleomagnetic analysis was done on specimens from 23 and 19 sites located in and around the Chipman Lake and Seabrook Lake complexes, respectively, using detailed alternating-field and thermal step demagnetization and isothermal remanent magnetization tests. Contact tests with adjacent Archean host rocks show that both complexes retain a primary characteristic remanence (ChRM). The Chipman Lake's ChRM is retained in 11 dikes with normal polarity and one dike with reversed polarity and at one site with normal polarity and one site with reversed polarity from the fenite alteration zone. Its ChRM gives a pole position at 186°E, 38°N (dp = 7°, dm = 11°), which corresponds to a Keweenawan age of 1098 ± 10 Ma, suggesting that younger K–Ar amphibole ages do not date emplacement. The ChRM of the host rock, the Chipman Lake diorite stock, gives a pole at 49°E, 51°N (dp = 8°, dm = 13°), showing that it is not part of the Keweenawan complex but may be a 2.45 Ga Matachewan intrusive. The Seabrook Lake complex's ChRM is found at six normal polarity sites from within the complex and at four normal and three reversed polarity sites from within the fenitized Archean granite and Matachewan diabase of the contact aureole. It gives a pole position at 180°E, 46°N (dp = 11°, dm = 17°), which corresponds to a Keweenawan age of 1103 ± 10 Ma, agreeing with K/Ar biotite ages. The paleomagnetic data indicate that no significant motion on the Kapuskasing Structural Zone occurred after emplacement of the complexes excluding minor vertical uplift of less than about 4 km, and that there were multiple polarity transitions of a symmetric Earth's magnetic field during Keweenawan time.

Les affleurements du complexe de Chipman Lake représentent une série de carbonatite et de dykes de roches mafiques alcalines apparentées, dans la sous-province de Wabigoon, de la province du lac Supérieur, tandis que les affleurements du complexe de Seabrook Lake appartiennent à un petit massif intrusif de syénite alcaline – carbonatite, dans la sous-province de l'Abitibi. L'étude paléomagnétique a porté sur des spécimens provenant de 23 sites à l'intérieur des deux complexes, et de 19 sites autour d'eux-mêmes, utilisant les méthodes détaillées de champ alternatif, de désaimantation par chauffage et des tests d'aimantation rémanente isotherme. Les tests du contact avec les roches encaissantes archéennes révèlent que les deux complexes retiennent une rémanence caractéristique (RMca) primaire. La RMca du complexe de Chipman Lake est retenue dans 11 dykes avec polarité normale et dans un dyke avec une polarité inverse, et dans la zone d'altération en fénite se trouve un site avec polarité normale et un site avec polarité inverse. Sa RMca fournit une position de pôle à 186°E, 38°N (dp = 7°, dm = 11°), ce qui correspond à un âge keweenawien de 1098 ± 10 Ma, impliquant que les âges plus jeunes obtenus par K–Ar sur amphiboles ne datent pas cet emplacement. La RMca de la roche encaissante, le petit massif intrusif de la diorite de Chipman Lake, fournit un pôle à 49°E, 51°N (dp = 8, dm = 13°), ce qui prouve qu'il n'est pas intégré au complexe de Keweenawan, mais qu'il pourrait correspondre à une intrusion de Matachewan datée de 2,45 Ga. La RMca du complexe de Seabrook Lake a été décelée à six sites de polarité normale à l'intérieur du complexe, et à quatre sites de polarité normale et trois autres de polarité inverse dans le granite archéen fénitisé et la diabase de Matachewan de l'auréole de contact. Il fournit une position de pôle à 180°E, 46°N (dp = 11°, dm = 17°) qui correspond à un âge keweenawien de 1103 ± 10 Ma, en accord avec les âges K/Ar sur biotites. Les données paléomagnétiques indiquent l'absence de mouvement important dans la zone structurale de Kapuskasing après la mise en place des complexes, sauf pour un soulèvement vertical mineur inférieur à environ 4 km, et que sont apparues des transitions de polarité multiples du champ magnétique terrestre symétrique durant la période du Keweenawan.

[Traduit par la rédaction]

Can. J. Earth Sci. **29**, 1215–1223 (1992)

Introduction

The Chipman Lake and Seabrook Lake carbonatite complexes are the only 1.1 Ga Keweenawan intrusives emplaced into the Archean rocks of the Wabigoon and Abitibi subprovinces, respectively, of the Superior Province in the Canadian Shield (Fig. 1). The former is the most northerly and the latter the most easterly known Keweenawan alkalic complexes that outcrop sufficiently to permit paleomagnetic study for geotectonic analysis.

Geology at Chipman Lake

The Chipman Lake complex comprises a cluster of carbonatite dikes and adjacent fenites at the south end of Chipman Lake next to the Big Bay – Ashberton Bay fault (Fig. 2). This fault extends southward to Lake Superior past the 1.1 Ga Killala and Coldwell alkalic complexes. Displacement on the fault at Chipman Lake is about 1 km and could reflect either horizontal or vertical motion (Sage 1985). The carbonatite dikes cut Archean rocks of the Wabigoon Subprovince which include older mafic to intermediate metavolcanic rocks, metamorphosed diorite and hornblendite of the Chipman Lake stock, and younger foliated granitic rocks with late massive pegmatitic and aplitic veins. The Chipman Lake stock has a strong circular aeromagnetic anomaly that led Satterly (1970) and Biczok (1976) to consider it to be part of the Keweenawan complex. The late Precambrian dikes of the complex include a few diabase dikes up to 15 m wide with northerly strikes that parallel the fault, lamprophyre dikes up to 0.3 m wide, and mafic dikes up to 0.7 m wide that are entirely altered to biotite

[1]Lithoprobe Publication 278.

Printed in Canada / Imprimé au Canada

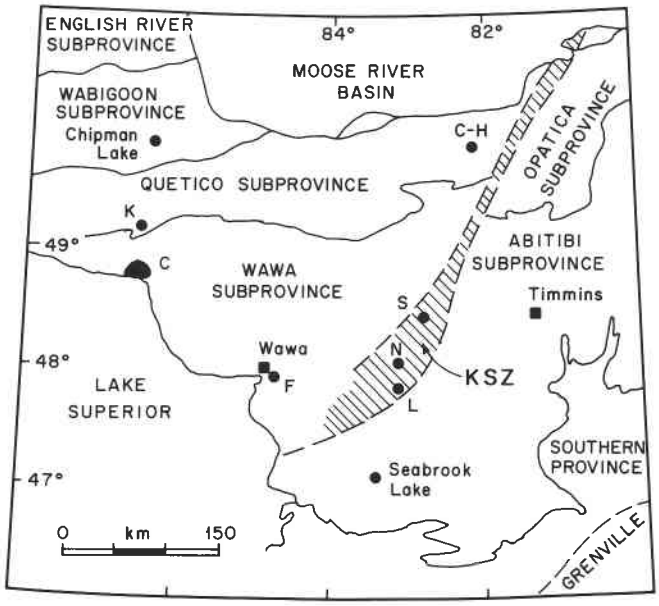


FIG. 1. Map of the Kapuskasing Structural Zone (KSZ) area of the Superior Province. Labelled carbonatite complexes, in addition to Seabrook Lake and Chipman Lake, include Clay–Howells (C–H), Coldwell (C), Firesand River (F), Killala (K), Lackner Lake (L), Nemegosenda (N), and Shenango (S).

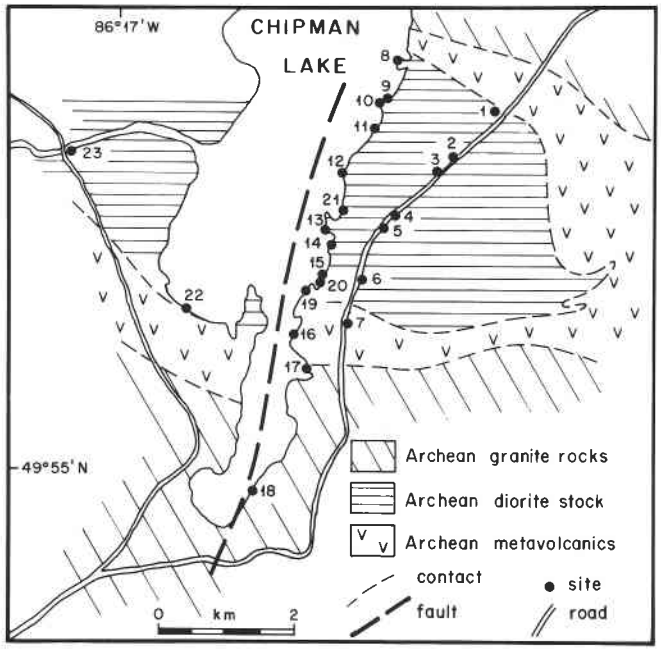


FIG. 2. Geology of the Chipman Lake carbonatite complex (after Sage 1985) and sampling locations.

(Sage 1985). The carbonatite dikes are generally less than 1 m wide and follow joints and fractures. They are composed of beforsite (dolomitic carbonatite) with substantial carbonate alteration of the adjacent wall rocks to fenite. Amphibole from the fenite gave a K–Ar age of 1022 ± 31 Ma (J.C. Baubron, personal communication, cited in Sage (1985)). Age relations between the late Precambrian dikes are uncertain; the dikes could represent more than one event.

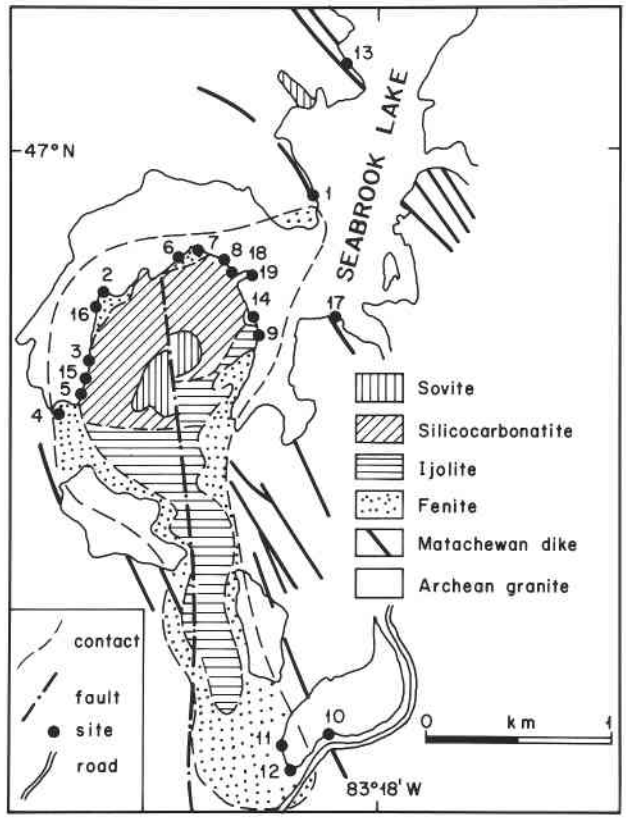


FIG. 3. Geology of the Seabrook Lake carbonatite complex (after Sage 1988) and sampling locations.

Geology at Seabrook Lake

The Seabrook Lake complex occurs at the intersection of several regional lineaments, some of which appear to be related to the Kapuskasing Structural Zone (KSZ) at its extreme southern end (Fig. 1). The complex has been mapped by Parsons (1961) and Sage (1988) (Fig. 3). It covers an area of 1.5 km^2 and has a tadpole shape. It comprises sovite, heterogeneous silicocarbonatite with mafic breccia, and minor syenite with a tail of ijolite that grades to malignite. The complex is emplaced into massive red-brown Archean granite of the Abitibi Subprovince that shows no evidence of regional recrystallization or metamorphism. The granites have been cut by north-northwest-trending Matachewan diabase dikes of Proterozoic age (2452^{+2}_{-3} Ma, Heaman 1988). These host rocks are brecciated and fenitized adjacent to the contact of the complex. Small dikes of carbonatite and lamprophyre cut the granite in the region around the complex. The tadpole shape of the complex can be clearly seen in its associated aeromagnetic anomaly (Ontario Department of Mines and Geological Survey of Canada 1963*a*, 1963*b*). The complex has been dated at 1119, 1117, and 1102 ± 50 Ma using the K–Ar method on biotite by Gittins *et al.* (1967) using current decay constants, giving a mean age of 1113 ± 36 Ma.

Experimental methods

At Chipman Lake five to seven 2.5 cm diameter cores were drilled at each of 23 sites and oriented *in situ* using a sun compass. Of the 23 sites, 9 are in the Archean Chipman Lake diorite stock and the remainder are in Proterozoic dikes of diabase (2), lamprophyre (2), biotite (2), and carbonatite (6) or

TABLE 1. Chipman Lakes complex mean remanence directions

Site	Geology[a]	Treatment[b] (°C, mT)	No. of cores and specimens[c]	D (°)	I (°)	α_{95} (°)	k	Co
1	Ad	530–585;10–40	6, 0, 13, 0	189	55	15	20	B
2	Ad	500–585;20–160	5, 0, 9, 0	197	−31	19	16	A
3	Ad	555–585;10–120	4, 1, 5, 1	214	−50	28	12	A
4	Ad	200–585;10–120	5, 0, 10, 0	161	79	12	45	B
5	Pl, 0.4, Ad	530–585;5–100	6, 0, 12, 0	287	46	7	102	C
6	Ad	530–585;20–80	5, 0, 6, 3	216	−21	10	56	A
7	Pd, 0.2, Av	530–585;20–60	5, 0, 9, 0	316	43	17	22	C
8	Pf	530–585;20–60	5, 0, 5, 4	97	−26	10	58	C
9	Pb, 0.1, Ad	200–500;10–160	6, 0, 12, 0	303	44	4	261	C
10	Pb, 0,1, Ad	200–550;10–50	5, 0, 11, 0	296	56	9	69	C
11	Pc, 0,3, Ad	200–555;0–160	5, 0, 9, 0	297	49	12	41	C
12	Ad	500–585;10–160	7, 0, 14, 0	201	−21	7	73	A
13	Pc, 0.4, Ad	450–530;20–140	6, 0, 15, 0	302	25	18	16	C
14	Pf	450–585;20–60	6, 0, 13, 0	276	64	12	30	C
15	Ad	200–585;15–140	6, 0, 13, 0	222	−42	9	54	A
16	Pc, 0.3, Av	530–570;30–50	4, 2, 3, 2	113	−48	15	37	C
17	Pl, 4.0, Ag	350–585;10–100	6, 0, 13, 0	284	58	10	46	C
18	Pc, 1.3, Ag	200–570;20–80	5, 0, 12, 0	334	60	19	17	C
19	Pd, 15, Ad	450–645;10–50	5, 0, 10, 0	295	56	10	58	C
20	Pc, 0.3, Ad	500–585;20–100	6, 0, 14, 0	303	65	7	90	C
21	Ad	500–585;20–140	5, 1, 9, 0	202	−18	13	35	A
22	Pc, 0.1, Ag	530–585;20–160	6, 0, 10, 0	293	30	7	91	C
23	Ad	450–570;30–140	4, 1, 3, 3	207	−48	11	71	A
Mean, component A			7	207.9	−33.3	11.8	27.2	
Mean, component B			2	182	68		19.9	
Mean, component C[d]			14	295.6	48.8	8.2	24.5	

NOTES: The site mean remanence vector, obtained by averaging core means, is given by its declination (D), inclination (I), radius of 95% confidence (α_{95}), its precision parameter (k) (Fisher 1953), and its assigned component group (Co).

[a]Site lithology and dike width in metres and its host rock if appropriate. Symbols for Archean rocks: Av, metavolcanics; Ag, granite; Ad, Chipman Lake diorite stock. Symbols for Proterozoic rocks: Pf, fenite; Pc, carbonatite dike; Pl, lamprophyre dike; Pb, biotite dike; Pd, diabase dike.

[b]Demagnetization temperature range and peak AF range of accepted specimen remanence vectors.

[c]Number of cores with accepted vectors, cores with unaccepted vectors, specimens with accepted end-point vectors, and specimens with accepted remagnetization-circle vectors, respectively. The value given for the mean component remanence vector is the number of sites included in the average.

[d]The mean directions for sites 8 and 16 have been reversed to calculate the C mean component.

in associated fenites (2) (Fig. 2; Table 1). Two or three specimens were sliced from each core. Their natural remanent magnetization (NRM) was measured using an automated CTF cryogenic magnetometer inside a shielded room after 1 month of storage to minimize the acquisition of viscous remanent magnetization (VRM) from the present Earth's magnetic field (PEMF). Two specimens from each site were demagnetized using an alternating field (AF) in 14 steps to 160 mT with a Sapphire Instrument SI-4 model AF demagnetizer. An additional two specimens from each site were thermally demagnetized in 10 steps to 615°C using a Schonstedt TSD-1 model thermal demagnetizer, with five additional steps for a few specimens to 690°C. The response of the remanence vector in the specimens from a given site was essentially the same for both AF and thermal step demagnetization, so the remaining specimens were AF demagnetized in four steps to best define their characteristic remanent magnetization (ChRM) direction over a broad coercivity spectrum of greater than 40 mT in most cases (Table 1). The ChRM directions have maximum angular deviations of less than 10° using principal component analysis, with most being less than 5° (Kirschvink 1980); these were combined using conventional tiered analysis to get the site mean directions (Fisher 1953). For a few sites the remagnetization circle method was used to estimate the remanence directions for some specimens, and these were combined with the ChRM directions to get the site means (Bailey and Halls 1984) (Table 1). Isothermal remanent magnetization (IRM) tests were run on eight representative specimens by magnetizing them in eight direct-field (DF) steps to 200 mT and then AF demagnetizing them in six steps to 40 mT to help characterize their magnetic mineralogy.

The analysis of the Seabrook Lake complex followed the same procedure. Eight of its 19 sites are in sovite (3), silicocarbonatite (1), mafic breccia (3), and ijolite (1). An additional seven sites are in its aureole of fenitized granite (2) or granite breccia (5) (Fig. 3; Table 2). The remaining four sites are in Matachewan dikes at different distances from the complex.

Results

Chipman Lake complex

Archean Chipman Lake stock

Step demagnetization of specimens from the Chipman Lake diorite stock reveals two ChRM components after the initial VRM from the PEMF is removed by about 20 mT or 350°C. At seven sites (2, 3, 6, 12, 15, 21, and 23) the ChRM direction is to the south-southwest and is inclined moderately upwards; this is termed the *A* magnetization component. In a few speci-

TABLE 2. Seabrook Lake complex mean remanence directions

Site	Geology	Treatment (°C; mT)	No. of cores and specimens	D (°)	I (°)	α_{95} (°)	k	Co
1	Amd	—;15–80	4, 1, 7, 0	122	−42	18	27	E2
2	Afbg	—;10–100	2, 3, 3, 1	306	60	—	49	E2
6	Afgb	—;10–100	3, 2, 3, 0	282	56	30	17	E2
8	Pso	—;10–40	3, 2, 5, 0	271	32	17	52	E1
9	Pi	—;30–60	2, 3, 3, 0	314	61	—	199	E1
10	Amd	500–580;10–100	4, 1, 7, 0	230	5	8	139	D
11	Afgb	—;10–80	3, 2, 3, 1	91	−21	17	51	E2
13	Amd	500–580;10–30	2, 0, 3, 0	33	33	—	35	D
13	Amd	450–580;10–30	4, 0, 7, 0	355	73	13	51	E2
14	Psc	400–560;20–130	3, 2, 6, 0	313	44	23	29	E1
15	Psc	300–850;20–160	6, 0, 9, 2	334	44	9	51	E1
16	Afgb	—;10–160	6, 0, 9, 2	154	−57	33	15	E2
17	Amd	—;30–160	2, 0, 4, 0	44	35	—	8	D
17	Amd	560–580;40–160	4, 0, 8, 0	325	17	10	81	E2
18	Pso	—;10–40	4, 1, 8, 0	333	45	15	37	E1
19	Pso	300–580;10–60	5, 0, 13, 0	314	25	14	31	E1
Mean, component D			3	42.8	21.4	37.8	11.6	
Mean, component E1			6	312.4	43.9	18.1	14.6	
Mean, component E2			7	304.9	50.0	21.8	8.6	
Mean, component E			13	308.7	47.2	12.9	11.2	

NOTES: Site lithology symbols for Archean rocks: Afgb, fenitized granite breccia; Amd, Matachewan diabase. Symbols for Proterozoic rocks: Pi, ijolite; Psc, silicocarbonatite; Pso, sovite. Other headings are explained in Table 1 notes. The mean directions for sites 1, 10, 11, and 16 have been reversed to calculate the D and E component mean directions.

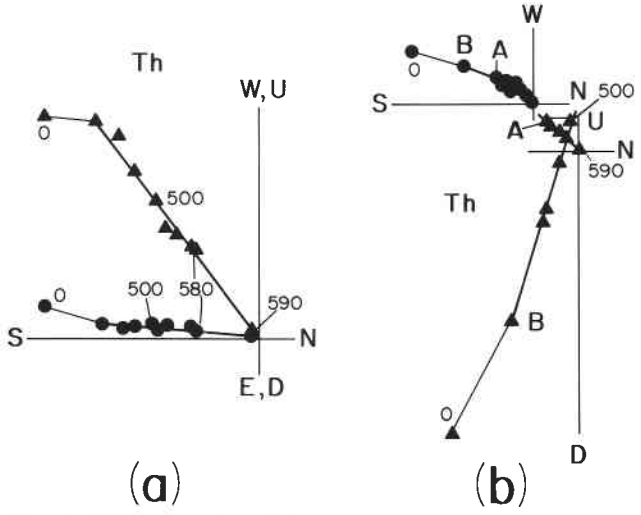


FIG. 4. Orthogonal plot of the remanent magnetization vector on thermal (Th) step demagnetization for a specimen from the Archean Chipman Lake diorite stock having (*a*) the A component with an NRM intensity of $J_0 = 9.6 \times 10^{-1}$ A/m, and (*b*) the B component, being removed before isolating the A component, with an NRM intensity of $J_0 = 6.9 \times 10^{-1}$ A/m. The circles denote the horizontal projection (N–S, E–W axes), and the triangles denote the vertical projection (U–D, N–S axes). The temperatures of some steps are shown in degrees Celsius.

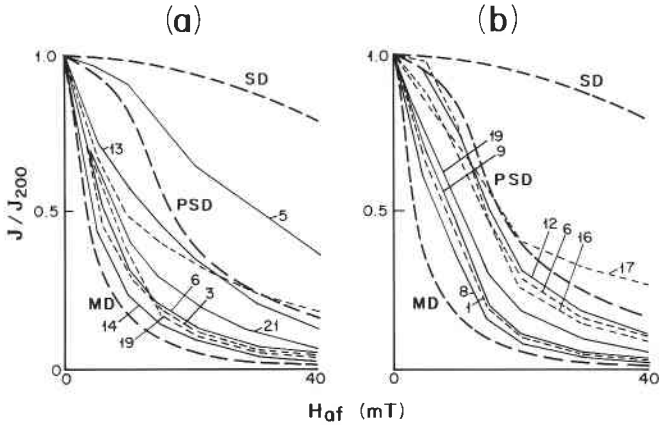


FIG. 5. Isothermal remanent magnetization results for test specimens from the (*a*) Chipman Lake and (*b*) Seabrook Lake complexes. The decrease is shown in the ratio of the remanence intensity to the intensity acquired in a direct field of 200 mT (J/J_{200}) upon alternating field demagnetization (H_{af}). Also shown are the site numbers for the specimens and type curves for single domain (SD), pseudosingle domain (PSD), and multidomain (MD) magnetite.

mens *A* resides in single-domain (SD) magnetite, as shown by the rapid decrease in remanence intensity between 585 and 610°C on thermal step demagnetization (Fig. 4*a*); however, in most specimens it resides in pseudosingle-domain (PSD) to multidomain (MD) magnetite crystals, as shown by the rapid rate of IRM acquisition and decay on AF demagnetization (Fig. 5*a*, sites 3, 6, and 21).

At two sites (1 and 4) the ChRM direction is to the south and is inclined steeply downwards; this is termed the B magnetization component. B resides in magnetite in a somewhat lower range of AF coercivities and thermal unblocking temperatures than the A component. In several specimens it is clear that the B component is being demagnetized before the A component is isolated at higher coercivities and unblocking temperatures (Fig. 4*b*). This indicates that the A component predates the B component because both are held by magnetite. The site mean directions for the A component are reasonably well grouped, whereas those for the two B-component sites are at about right

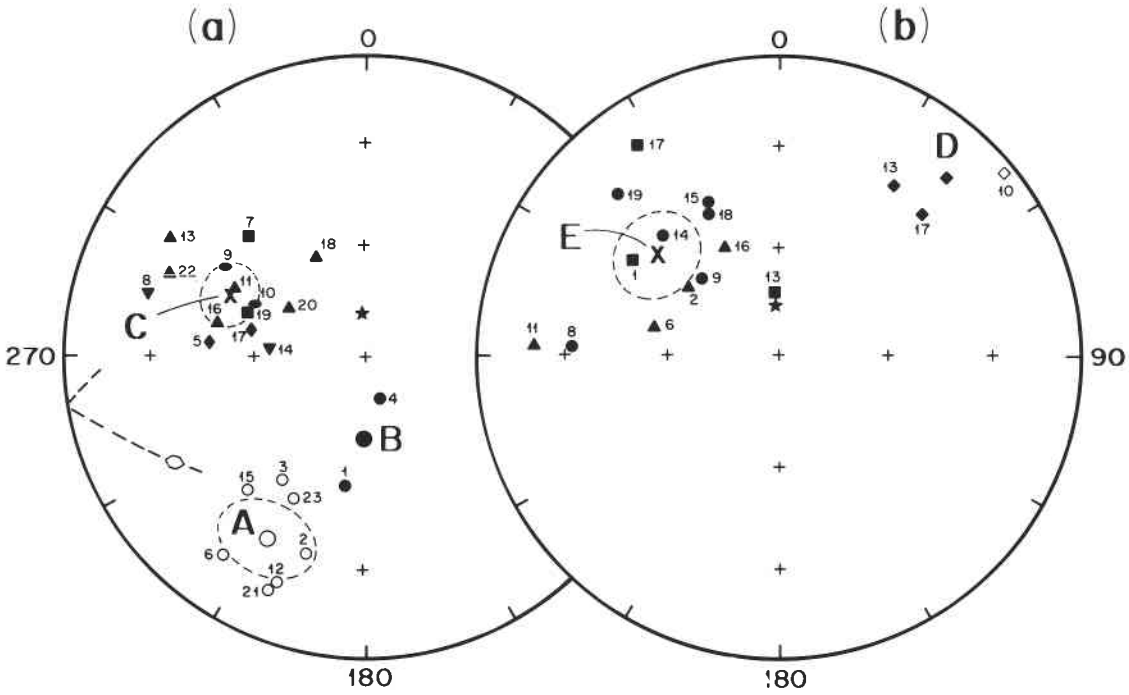


FIG. 6. Equal-area stereograms for the site mean directions from the (*a*) Chipman Lake and (*b*) Seabrook Lake complexes (Tables 1, 2). Closed (open) symbols represent positive down (negative up) directions. The larger symbols denote the component mean directions and are surrounded by their cones of 95% confidence where well defined. The star is the present Earth's magnetic field direction. In (*a*) the lithologies are Archean diorite (circles) and Proterozoic dikes of biotite (oval), carbonatite (triangles), diabase (squares) and lamprophyre (diamonds) with adjacent fenite alteration (inverted triangles). The hexagon is a ChRM direction from a diorite specimen in the baked contact of the dike at site 20. In (*b*) the lithologies are Archean Matachewan diabase dikes (diamonds), Proterozoic carbonatite complex (circles) and its remagnetized aureole of granite (triangles) and diabase (squares). The mean directions are shown in their antiparallel positions for sites 8 and 16 in (*a*) , and 1, 10, 11, and 16 in (*b*).

angles to A but are relatively close to each other (Fig. 6*a*; Table 1).

Proterozoic intrusives

The Proterozoic dike specimens all give mean ChRM directions to the west-northwest at moderate downward inclinations (Figs. 7*a*–7*c*) except for specimens from two sites (8 and 16) which are antiparallel (Fig. 7*d*). This is termed the C magnetization component. A range of demagnetization behaviour is expected given the range of rock types represented at the 14 sites. One generalization is that the C component is easily isolated in these rocks after the initial VRM is removed by about 10–15 mT and 350–450°C on AF and thermal step demagnetization, respectively. A second generalization is that the ChRM directions of the lamprophyre and biotite dike sites tend to be somewhat better defined than in the fenite alteration zone or carbonatite and diabase dike sites, using the radii of their cones of 95% confidence (α_{95}) and precision parameters (k) as a measure (Fisher 1953). Nevertheless, the C component is clearly defined at all 14 sites.

In almost all specimens magnetite with unblocking temperatures of less than 600°C is the significant carrier of C remanence (Fig. 7*d*). There are, however, a few specimens that retain a small C component in hematite with unblocking temperatures over 600°C (Fig. 7*b*). The IRM tests show that the C component resides dominantly in SD to PSD magnetite in the fine-grained lamprophyre dikes (Fig. 5*a*, site 5), in PSD to MD magnetite in the carbonatite veins (site 13), and in MD magnetite in the fenites and diabase dikes (sites 14 and 19).

In most cases the dikes and veins proved to be too narrow to provide a diagnostic paleomagnetic contact test (Everitt and Clegg 1962) showing the baked zone of the host rock to have the dike's ChRM direction. In one test between a carbonatite dike (site 20) cutting Archean diorite (site 15), a specimen with an intermediate direction between the two site means was found in the baked zone (hexagon, Fig. 6*a*). However, the preservation of the C component in all Proterozoic dikes and the fenitized diorite, when contrasted with the distinctly different directions of the A or B component in the Archean host diorite, does provide a reasonable contact test to show that C is younger than either A or B.

When the site mean direction for each of the Proterozoic rock types is compared with the mean of the Proterozoic group as a whole, there is no evident bias; the mean of each rock type falls within the cone of 95% confidence of the overall mean. This suggests that all the Proterozoic rock units were emplaced at about the same time, and not as two or more discrete events separated by a few tens of millions of years. Obviously, the two sites with reversed magnetization (8 and 16) record a reversed event that differs in time from the normal event. However, the comparison test of McFadden and Lowes (1981) shows that the two populations are not significantly different in direction, and therefore in time, within the limits of paleomagnetic resolution. The implication is that the normal and reversed events record a reversal in the Earth's magnetic field during the several million year period of emplacement and cooling of the Chipman Lake complex. The transition could be normal-to-reversed or reversed-to-normal.

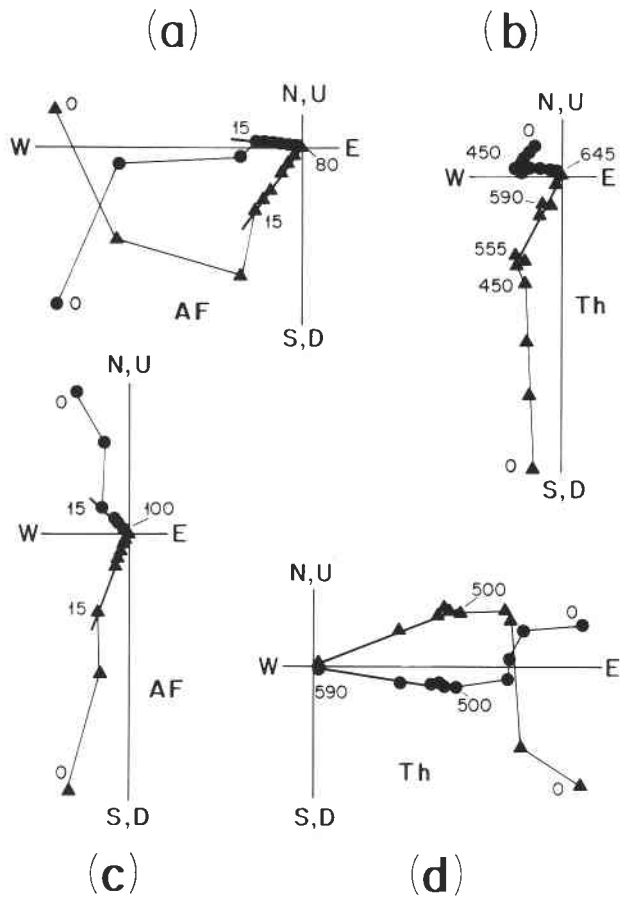


FIG. 7. Orthogonal plots for specimens from Chipman Lake complex in dikes with the normal C component of (*a*) lamprophyre, $J_0 = 3.75 \times 10^{-1}$ A/m; (*b*) diabase, $J_0 = 11.5 \times 10^{-1}$ A/m; and (*c*) carbonatite, $J = 10.9 \times 10^{-1}$ A/m; and (*d*) with the reversed C component of carbonatite, $J_0 = 3.7 \times 10^{-3}$ A/m. Conventions as in Fig. 4 except that alternating-field (AF) demagnetization step intensities are in milliTesla.

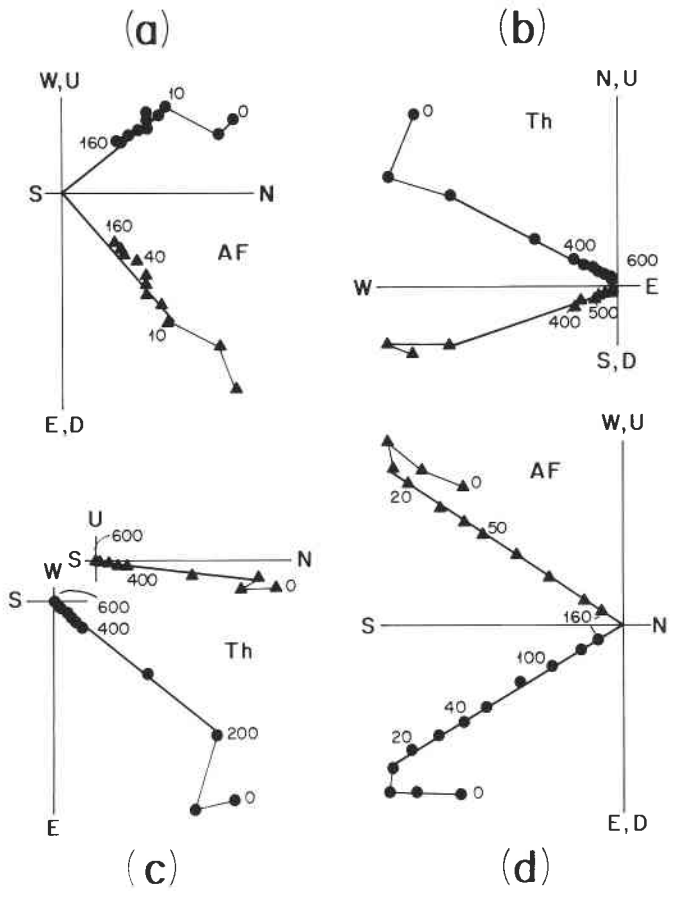


FIG. 8. Orthogonal plots for specimens from the Seabrook Lake complex showing the normal E component in (*a*) silicocarbonatite, $J_0 = 8.3 \times 10^{-4}$ A/m, and (*b*) sovite, $J_0 = 9.7 \times 10^{-4}$ A/m; (*c*) the normal D component in Matachewan diabase, $J_0 = 2.5 \times 10^{-1}$ A/m; and (*d*) the reversed E component in fenitized granite breccia from the contact aureole, $J_0 = 11.6 \times 10^{-1}$ A/m. Conventions as in Figs. 4 and 7.

Seabrook Lake complex

Carbonatite complex

After removal of the initial VRM from the PEMF by about 300°C on thermal step demagnetization or by about 10–20 mT on AF step demagnetization, a ChRM was found in some or all specimens from six of eight sites in the complex (Fig. 8*a*). It is directed to the northwest, is inclined at an intermediate angle downwards, and is termed the E1 component (Table 2). Almost all sovite and silicocarbonatite specimens (sites 8,15,18, and 19) gave consistent E1 directions. The ijolite site (9) and one mafic breccia site (14) gave some specimens with useful E1 magnetization components, whereas two of the mafic breccia sites (3 and 5) gave no specimens with coherent ChRM's. In general, AF step demagnetization proved more efficacious than thermal step demagnetization. However, thermal step demagnetization does show clearly that magnetite is the sole significant remanence carrier at these sites (Fig. 8*b*). The IRM acquisition and demagnetization curves show that the magnetite in the complex ranges from PSD to MD in behaviour (Fig. 5*b*, sites 8, 9, 12, and 19). Taken together the six E1 site mean remanence directions from the Seabrook Lake complex form a moderately coherent cluster (Table 2; Fig 6*b*).

Archean host rocks

The fenitized granite at two sites (4 and 12) and fenitized granite breccia at one site (7) failed to retain a coherent ChRM. Some specimens from each of the four other sites of fenitized granite breccia do, however, retain a ChRM directed to the northwest at an intermediate angle downwards (sites 2 and 6) or to the antiparallel position (sites 11 and 16) (Fig. 8*d*). This is termed the E2 remanent magnetization component. The fenitized granite breccia is next to the contact with the carbonatite complex and is surrounded in turn by the fenitized granite. Therefore the breccia sites are apt to be both more highly fenitized and contact metamorphosed to realign their ChRM direction to the complex's E1 direction. The remanence data for the fenitized granite breccias come mostly from AF step demagnetization because thermal demagnetization proved ineffective. Nevertheless there is no evidence that hematite is the ChRM carrier. The IRM curves (Fig. 5*b*, sites 6 and 16) are characteristic of PSD magnetite with minor MD magnetite, suggesting strongly that the E2 magnetization is dominantly a thermal overprint acquired over a sufficiently long period of time to permit the Earth's magnetic field to undergo a polarity reversal.

Specimens were collected at four sites in Matachewan diabase dikes that crop out at distances of about 50 (site 1), 200

TABLE 3. Palaeomagnetic pole positions

Component	Unit	N^a	Pole position		Radii of $P_{0.05}$ oval	
			Long. E (°)	Lat. N (°)	dp (°)	dm (°)
		CHIPMAN LAKE				
A	Diorite stock, primary	7	48.6	51.2	7.6	13.4
B	Diorite stock, secondary	2	272.6	11.0	—	—
C	Carbonatite complex	14	186.1	38.4	7.1	10.8
		SEABROOK LAKE				
D	Matachewan diabase dikes	3	37.4	39.2	—	—
E	Carbonatite complex	13	180.2	46.3	10.8	16.7

[a]Number of site mean remanence vectors averaged to get the pole position.

(site 10), 250 (site 13), and 300 m (site 17) from the nearest outcrop of carbonatite or fenitized granite. The specimens from the closest site (1) give a well-defined E2 reversed-polarity ChRM direction. The IRM acquisition and decay curve for a specimen from site 1 is typical of MD magnetite (Fig. 5*b*). Two more distant dikes (sites 13 and 17) give two ChRM directions. One is the normal E2 direction and the other is directed to the north-northeast with a shallow downward inclination (Fig. 8*c*). The latter is termed the D magnetization component, and it has the same direction as the primary normal-polarity remanence isolated in Matachewan dikes by Bates and Halls (1990) along their Rocky Island Lake profile. This profile is located about 15 km south of Seabrook Lake, crossing the southward extension of the Matachewan dike swarm. The dikes with normal polarity on the profile give a mean remanence direction of $D = 19°$, $I = 21°$ ($\alpha_{95} = 19°$) (Bates and Halls 1990) that is statistically the same as the D ChRM direction. The fourth dike (site 10) gives a well-defined, reversed-polarity D ChRM direction that is concordant with reversed-polarity dikes along the Rocky Island Lake profile; they have a mean direction of $D = 209°$, $I = -6°$ ($\alpha_{95} = 5°$). The thermal demagnetization data show that magnetite is the dominant remanence carrier in the Matachewan dikes. The IRM curves for the specimen from site 17 are typical of PSD magnetite with, perhaps, a minor hematite contribution (Fig. 5*b*, site 17; Table 2). These results suggest that a thermal event was responsible for resetting the D to E2 ChRM in these rocks.

The site mean E2 magnetization direction from the Archean fenitized granite breccia (Fig. 7*b*, sites 2, 6, 11, and 16) is statistically indistinguishable from Matachewan dikes' (sites 1, 13, and 17) E2 direction. Similarly the E2 population is statistically indistinguishable from the E1 population of the complex itself. Therefore the E1 and E2 populations have been combined to compute the mean E direction for the Seabrook Lake complex (Table 2). The results for the Seabrook Lake study indicate that Matachewan dikes with both normal and reversed primary D remanence were present as Archean host rocks. In the Keweenawan, the carbonatite complex was emplaced during a normal polarity period, acquiring its normal-polarity E magnetization. During emplacement and subsequent cooling, the Archean remanent magnetization of the fenitized granite breccias and closest Matachewan dike were overprinted with the E ChRM of both normal and reversed polarity, implying that the normal event was followed by a reversed event. The more remote Matachewan dikes were either partially remagnetized with a normal E ChRM by the peak thermal pulse during emplacement or were not remagnetized to retain their original D magnetization. This combination of results provides an intrusive-scale contact test to affirm that the Seabrook Lake complex retains a primary remanence (Everitt and Clegg 1962).

Discussion

The paleomagnetic pole position for the A remanent magnetization component of the Chipman Lake diorite stock is at 49°E, 51°N (Table 3). This pole is close to the late Archean apparent polar wander path (APWP) for the Superior Province of Vandall and Symons (1990) at the well-defined 2454 Ma position of the Matachewan diabase dike swarm. The Archean terrane around the Chipman Lake complex is intruded by the Matachewan dike swarm, and it gives a statistically indistinguishable remanence direction or overall pole (Bates and Halls 1991). Presumably the diorite stock was either metamorphosed at this time or is about coeval with the dike swarm. Radiometric age dating of the stock is required to assess this correlation.

The B component postdates the A component in the Chipman Lake stock, but probably predates the C component found in the Keweenawan dikes, as it was not found at any of the dike sites. B gives a poorly defined pole at 273°E, 11°N that, speculatively, approximately aligns with the 1850 Ma position on the Superior Province APWP. If true then it could reflect Proterozoic uplift from compression during the Hudsonian orogeny along with uplift of the KSZ (Symons and Vandall 1990).

The C component, found in the Keweenawan alkalic dikes at Chipman Lake, gives a reasonably well-defined pole at 186°E, 38°N (Table 3; Fig. 9), is shown to be primary by a contact test, and differs significantly from the A component of the diorite. Therefore Sage (1985) is correct to dispute Satterly's (1970) and Biczok's (1976) contention that the diorite is part of the alkalic complex. The location of the pole on the APWP indicates that the complex was emplaced at 1098 ± 10 Ma and that the K−Ar amphibole ages on adjacent alteration of 1022 ± 31 and 1029 ± 31 Ma, reported by J.C. Baubron (cited in Sage 1985), do not date emplacement.

The D remanence component, residing in Archean Matachewan dikes some hundreds of metres from the Seabrook Lake complex, gives a poorly defined pole at 37°E, 39°N. This pole is concordant with that obtained by Bates and Halls (1990) from Matachewan dikes a few kilometres to the south, indicating a primary magnetization acquired at about 2454 Ma.

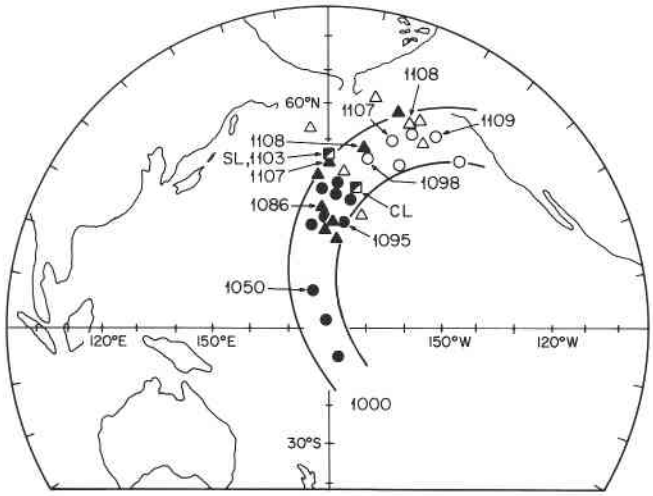


FIG. 9. The North American apparent polar wander path for Keweenawan time from Lewchuk and Symons (1990*b*, Fig. 8), showing the locations of the Chipman Lake complex pole from the C component (CL) and the Seabrook Lake complex pole from the E component (SL). Closed, partially closed, and open symbols denote normal, mixed and reversed polarity poles, respectively.

The E remanence component is found in the Seabrook Lake complex, in its aureole of fenetized Archean granite, and in adjacent Matachewan dikes. Data from these specimens provide a contact test that shows that the E component is primary. Its pole at 180°E, 46°N lies directly on the Keweenawan APWP (Fig. 9), giving a paleomagnetic age of about 1103 ± 10 Ma that agrees with its mean K–Ar biotite age of 1113 ± 36 Ma (Gittins *et al.* 1967). The paleomagnetic and radiometric results imply that the complex cooled quickly after emplacement, with no more than a very few kilometres of subsequent uplift.

Both the Chipman Lake C pole and the Seabrook Lake E pole fall directly on the Keweenawan APWP, implying that neither has undergone significant tilt or rotation since emplacement. This is consistent with the vertical attitudes of the Keweenawan dikes at Chipman Lake and of the Matachewan dikes at Seabrook Lake and indicates that the tadpole shape of the latter complex is not caused by tilting.

The presence of the reversed-polarity C component at a carbonatite dike site and a fenite alteration zone site in the Chipman Lake complex with its predominantly normal C magnetization and of the reversed-polarity E component at three sites in the contact zone of the Seabrook Lake complex with its predominantly normal E magnetization has important consequences. Both the C and E poles fall near the boundary on the APWP between predominantly reversed poles prior to about 1100 Ma and predominantly normal poles after 1100 Ma. Halls and Pesonen (1982) have proposed the use of this superchron polarity transition for the regional correlation of Keweenawan strata throughout the Midcontinental Rift System. Given the dual polarities found in the Chipman Lake complex and apparent normal-to-reversed polarity sequence found in the Seabrook complex, it is unlikely that they are recording the superchron transition. Rather these poles could be marking the transition from normal episode II to reversed episode III found in the Coldwell complex (Lewchuk and Symons 1990*b*), the last normal-to-reverse transition found in the Maimanse Point lava section (Palmer 1970; Robertson 1973), or one or two entirely different polarity transitions. The superchron model fails to account for (*i*) the locations of normal poles for the Lackner Lake complex (Symons 1989*a*), Nemegosenda complex (Symons and Garber 1974), Coldwell complex, and Maimanse Point lava section in the "reversed superchron" of Halls and Pesonen (1982); (*ii*) the poles for the apparently normal-to-reversed transition at Seabrook Lake; and (*iii*) the reversed pole from the Shenango complex (Dunlop *et al.* 1988) in the "normal superchron." A more likely model is one of predominantly reversed polarity from about 1120 to 1100 Ma, followed by predominantly normal polarity from about 1100 to 1080 Ma, with several reversals throughout the 40 Ma period.

In addition the Chipman Lake and Seabrook Lake results do not support the hypothesis of Pesonen and Nevanlinna (1981) that the Earth's magnetic field in Keweenawan time was asymmetric. This hypothesis requires that reversed-polarity sites should indicate higher paleolatitudes than normal sites, whereas the converse is true for both the Chipman Lake and Seabrook Lake sites.

Finally the Chipman Lake C pole from the Wabigoon subprovince and the Seabrook Lake E pole from the Abitibi subprovince join a growing list of concordant poles from the KSZ region. This list includes the Clay–Howells complex in the Quetico subprovince (Lewchuk and Symons 1990*a*), the Firesand River and Coldwell complexes in the Wawa subprovince (Symons 1989*b*; Lewchuk and Symons 1990*b*), and the Nemegosenda and Lackner Lake complexes in the KSZ itself (Symons and Garber 1974; Symons 1989*a*). This suggests that no statistically significant geotectonic motion has taken place in the KSZ region since the start of Keweenawan time. The Seabrook Lake result is particularly important because it is the first from east of the Ivanhoe Lake cataclastic zone. This zone defines the east side of the KSZ and is the listric thrust fault on which the KSZ was uplifted (Percival and Card 1983). The concordance of these poles to the APWP implies that only vertical uplift of the KSZ has occurred since the start of the Keweenawan. It also implies that the regional "reversed ess-flexure" of the Superior Province in the KSZ region, interpreted by Bates and Halls (1991) from the geometry and paleomagnetic character of the Matachewan dike swarm, must predate Keweenawan time.

Conclusions

The Chipman Lake and Seabrook Lake alkalic syenite–carbonatite complexes both retain a primary Keweenawan ChRM that is confirmed by contact tests. At Chipman Lake this ChRM is bipolar, recording normal and reversed remanence in both its dikes and fenite alteration zones. Its paleomagnetic pole position is concordant with the Keweenawan APWP at 1098 ± 10 Ma. The host rocks of the Chipman Lake diorite stock give a pole position that is significantly different and points to a Matachewan age of about 2.54 Ga. The ChRM of the Seabrook Lake complex is normal in polarity; however, ChRM's of both normal and reversed polarity are found in its contact aureole, favouring a sequence from normal to reversed polarity in the Earth's field. The pole position for the complex falls on the Keweenawan APWP at about 1103 ± 10 Ma, in agreement with available K–Ar radiometric age dates. Matachewan dikes outside the contact aureole give concordant normal and reversed Matachewan ChRM directions. The locations of the bipolar primary poles

for the Chipman Lake and Seabrook Lake complexes permit only simple vertical uplift since emplacement. They also add to a growing list of Keweenawan paleomagnetic data that favour (*i*) no significant rotation or translation of the KSZ since the Keweenawan; and (*ii*) multiple polarity transitions of a symmetric Earth's magnetic field during Keweenawan time.

Acknowledgments

The author thanks M. T. Lewchuk, J. Ceman, W. Banwell, and D. Tilden for assistance in sample collecting, J. Kulhanek for help with the measurements, and the Natural Sciences and Engineering Research Council of Canada for funding this research. The manuscript was greatly improved by the helpful comments of H.C. Halls and H.C. Palmer whom I thank.

Bailey, R. C., and Halls, H. C. 1984. Estimate of confidence in paleomagnetic directions derived from mixed remagnetization circle and direct observational data. Journal of Geophysics, **54**: 174–182.

Bates, M. P., and Halls, H. C. 1990. Regional variation in paleomagnetic polarity of the Matachewan dyke swarm related to the Kapuskasing Structural Zone, Ontario. Canadian Journal of Earth Sciences, **27**: 200–211.

Bates, M. P., and Halls, H. C. 1991. Broad-scale Proterozoic deformation of the central Superior Province revealed by paleomagnetism of the 2.45 Matachewan dyke swarm. Canadian Journal of Earth Sciences, **28**: 1780–1796.

Biczok, J. 1976. The petrography and geochemistry of the Chipman Lake alkaline-carbonatite complex. B.Sc. thesis, Department of Geology, Lakehead University, Thunder Bay, Ont.

Dunlop, D. J., Constanzo-Alvarez, V., and Pesonen, L. J. 1988. Keweenawan age magnetization of carbonatite complexes and remagnetization of the Shawmere anorthosite, KSZ. Proceedings, Lithoprobe Workshop on the Kapuskasing Structural Zone II, Toronto, Ont., pp.191–204.

Everitt, C. W. F., and Clegg, J. A. 1962. A field test of paleomagnetic stability. Geophysical Journal of the Royal Astronomical Society, **6**: 312–319.

Fisher, R. A. 1953. Dispersion on a sphere. Proceedings of the Royal Society of London, Series A, **217**: 295–305.

Gittins, J., MacIntyre, R. M., and York, D. 1967. The ages of carbonatite complexes in eastern Canada. Canadian Journal of Earth Sciences, **4**: 651–655.

Halls, H. C., and Pesonen, L. J. 1982. Paleomagnetism of Keweenawan rocks. *In* Geology and tectonics of the Lake Superior basin. *Edited by* R. J. Wold and W. J. Hinze. Geological Society of America, Memoir 156, pp. 173–201.

Heaman, L. M. 1988. A precise U-Pb zircon age for a Hearst dyke. Geological Association of Canada, Program with Abstracts, **13**: A53.

Kirschvink, J. L. 1980. The least-squares line and plane and the analysis of paleomagnetic data. Geophysical Journal of the Royal Astronomical Society, **62**: 699–718.

Lewchuk, M. T., and Symons, D. T. A. 1990*a*. Paleomagnetism of the Clay–Howells carbonatite complex: constraints on Proterozoic motion in the Kapuskasing Structural Zone, Superior Province, Canada. Tectonophysics, **172**: 67–75.

Lewchuk, M. T., and Symons, D. T. A. 1990*b*. Paleomagnetism of the Late Precambrian Coldwell complex, Ontario, Canada. Tectonophysics, **184**: 73–86.

McFadden, R. L., and Lowes, F. J. 1981. The discrimination of mean directions drawn from Fisher distributions. Geophysical Journal of the Royal Astronomical Society, **61**: 19–33.

Ontario Department of Mines and Geological Survey of Canada. 1963*a*. Marisca Lake. Ontario Department of Mines and Geological Survey of Canada, Aeromagnetic Map 2229G, scale 1 : 63 360.

Ontario Department of Mines and Geological Survey of Canada. 1963*b*. Mashagama Lake. Ontario Department of Mines and Geological Survey of Canada, Aeromagnetic Map 2228G, scale 1 : 63 360.

Palmer, H. C. 1970. Paleomagnetism and correlation of some middle Keweenawan rocks, Lake Superior. Canadian Journal of Earth Sciences, **7**: 1410–1436.

Parsons, G. E. 1961. Niobium bearing complexes east of Lake Superior, Algoma and Sudbury districts, Ontario. Ontario Department of Mines, Geological Report 3.

Percival, J. A., and Card, K. D. 1983. Archean crust as revealed in the Kapuskasing uplift, Superior Province, Canada. Geology, **11**: 323–326.

Pesonen, L. J., and Nevanlinna, H. 1981. Late Precambrian Keweenawan asymmetric reversals. Nature (London), **194**: 436–439.

Robertson, W. A. 1973. Pole positions from the Mamainse Point lavas and their bearing on a Keweenawan pole path and polarity sequence. Canadian Journal of Earth Sciences, **10**: 1541–1555.

Sage, R. P. 1985. Geology of carbonatite–alkalic rock complexes in Ontario: Chipman Lake area. Ontario Geological Survey, Study 44.

Sage, R. P. 1988. Geology of carbonatite–alkalic rock complexes in Ontario: Seabrook Lake carbonatite complex. Ontario Geological Survey, Study 31.

Satterly, J. 1970. Aeromagnetic maps of carbonatite–alkalic complexes in Ontario. Ontario Department of Mines and Northern Affairs, Preliminary Map P.452 (revised).

Symons, D. T. A. 1989*a*. Paleomagnetism of the 1.1 Ga Lackner Lake complex and tectonics of the Kapuskasing Structural Zone. Canadian Journal of Earth Sciences, **26**: 1778–1783.

Symons, D. T. A. 1989*b*. Age of the Firesand River carbonatite complex from paleomagnetism. Canadian Journal of Earth Sciences, **26**: 2401–2405.

Symons, D. T. A., and Garber, J. 1974. Pole position for the Neohelikian Nemegosenda carbonatite complex, Ontario. Canadian Journal of Earth Sciences, **11**: 341–345.

Symons, D. T. A., and Vandall, T. A. 1990. Paleomagnetic evidence for Proterozoic tectonism in the Kapuskasing Structural Zone, Ontario. Journal of Geophysical Research, **95**: 19 199 – 19 211.

Vandall, T. A., and Symons, D. T. A. 1990. Paleomagnetism of Archean granites and Matachewan dikes in the Wawa subprovince, Ontario: reevaluation of the Archean apparent polar wander path. Canadian Journal of Earth Sciences, **27**: 1031–1039.